Maintenance Schedule

Dieselmotor
12V1163TB93
16V1163TB93
20V1163TB93
Anwendungsgruppe 1DS

M050674/08E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
12V1163TB93	370 kW/cyl.	1DS; continuous operation, variable, low load
16V1163TB93	370 kW/cyl.	1DS; continuous operation, variable, low load
20V1163TB93	370 kW/cyl.	1DS; continuous operation, variable, low load

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to §15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

- QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
- QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
- QL3: Maintenance work which requires partial disassembly of the product.
- QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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1.3 Load profile

Load profile

Power	100%	70%	<10%
Corresponding operating time	10%	70%	20%

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1.4 Maintenance schedule matrix

0 to 14,500 Operating hours

Task	Limit	Operating hours [h]																												
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
Engine																															
W0500	-	X																													
W0501	-	X																													
W0502	-	X																													
W0503	-	X																													
W0505	-	X																													
W0506	-	X																													
W0507	-	X																													
W0508	-	X																													
W1100	2 a																														
W1096	2 a																														
W1024	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1080	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1001	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1053	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1002	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1005	2 a							X						X						X						X					
W1026	2 a							X						X						X						X					
W1017	9 a							X						X						X						X					
W1142	9 a							X						X						X						X					
W1143	9 a							X						X						X						X					
W1038	18 a													X												X					
W2012	9 a													X												X					
W2009	9 a													X												X					
W2002	9 a													X												X					
W2036	9 a													X												X					
W2038	9 a													X												X					
W2041	9 a													X												X					
W2043	9 a													X												X					
W2065	9 a													X												X					
W2068	9 a													X												X					
W2069	9 a													X												X					
W2084	9 a													X												X					
W2088	9 a													X												X					
W2095	9 a													X												X					
W2062	9 a													X												X					
W2110	9 a													X												X					
W2111	9 a													X												X					
W1063	9 a													X												X					
W1212	9 a													X												X					
W1136	9 a																			X											
W1137	9 a																			X											

w = weeks
m = months
a = years

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